Judiciary System Database - Query Suggestions

IT214 Database Management System Project

SCHEMA ANALYSIS

Based on the provided schema, here's the reconstructed entity relationship:

Core Entities:

1. **CASE** (curr_status table)
   - case*no (PK), type, Category, Jurisdiction*Type, Name, Description, curr_status
2. **COURT** (Phone_No table - appears to be Court entity)
   - ID (PK), Location, Level, Pin*Code, Email, Phone*No
3. **JUDGE/PERSONNEL** (Experience table)
   - ID (PK), Name, Gender, DOB, Email, Status, Phone_No, Address, Position, Qualification, Experience
4. **LAWYER/ADVOCATE** (BAR*Registration*No table)
   - ID (PK), Name, Type, Link, BAR*Registration*No
5. **PARTY/ENTITY** (Gender table - appears to be Party/Entity)
   - ID (PK), Name, Type, Address, Phone, Email, Gender

Relationship Tables:

1. **HEARING** (Status table)
   - Hearing*No (PK), case*no (PK, FK), Type, Status
2. **COURT-CASE Assignment** (case_no table)
   - court*id (PK, FK), case*no (PK, FK)

3. **JUDGE-COURT Assignment** (court_id table)
    - judge*id (PK, FK), court*id (PK, FK)
4. **PANEL** (panel_id table)
    - case*no (PK, FK), hearing*no (PK, FK), panel_id (PK, FK)
5. **ORGANIZATION** (dissolve_date table)
    - ID (PK), type, form*date, dissolve*date
6. **CASE STATUS** (dormant_reason table)
    - ID (PK), status, curr*stage, is*dormant, dormant_reason
7. **PRECEDENT** (precedent*case*no table)
    - case*no (PK, FK), precedent*case_no (PK, FK)

SUGGESTED QUERIES

BASIC QUERIES (Similar to Lab04 - Level 1)

Query 1: Active Cases by Category

Description: List all cases that are currently active and belong to 'Criminal' category

Relational Algebra:

```
σ_(curr_status='Active' ∧ Category='Criminal')(CASE)
```

SQL:

```
SELECT case_no, Name, Type, Jurisdiction_Type, Description
FROM curr_status
WHERE curr_status = 'Active' AND Category = 'Criminal';
```

Query 2: Courts in Specific Location

Description: List all courts (ID, Location, Level) located in a specific city (e.g., 'Ahmedabad')

Relational Algebra:

```
π_(ID, Location, Level, Email)(σ_(Location LIKE '%Ahmedabad%')(COURT))
```

SQL:

```
SELECT ID, Location, Level, Email, Phone_No
FROM Phone_No
WHERE Location LIKE '%Ahmedabad%';
```

Query 3: Judges with Specific Qualification

Description: List all judges (ID, Name, Position) who have 'LLM' qualification

Relational Algebra:

```
π_(ID, Name, Position, Qualification)(σ_(Qualification='LLM' ∧ Position='Judge')(JUDGE))
```

SQL:

```
SELECT ID, Name, Position, Qualification, Experience
FROM Experience
WHERE Qualification = 'LLM' AND Position = 'Judge';
```

Query 4: Hearings on Specific Date for a Case

Description: List all hearings (Hearing_No, Type, Status) for case 'CR/2024/001'

Relational Algebra:

```
π_(Hearing_No, Type, Status)(σ_(case_no='CR/2024/001')(HEARING))
```

SQL:

```
SELECT Hearing_No, Type, Status
FROM Status
WHERE case_no = 'CR/2024/001';
```

INTERMEDIATE QUERIES (Multi-table Joins - Level 2)

Query 5: Cases with Their Assigned Courts

Description: List all cases (case_no, case name, court location) with their assigned court details

Relational Algebra:

```
π_(c.case_no, c.Name, ct.Location, ct.Level)(
  (ρ_c(CASE) ⋈_(c.case_no = cc.case_no) ρ_cc(COURT_CASE))
  ⋈_(cc.court_id = ct.ID) ρ_ct(COURT)
)
```

SQL:

```
SELECT cs.case_no, cs.Name AS case_name, cs.Category,
       c.Location AS court_location, c.Level AS court_level
FROM curr_status cs
JOIN case_no cn ON cs.case_no = cn.case_no
JOIN Phone_No c ON cn.court_id = c.ID;
```

Query 6: Judges Assigned to Specific Court

Description: List all judges (ID, Name, Experience) working in court with ID = 5

Relational Algebra:

```
π_(j.ID, j.Name, j.Experience, j.Position)(
  (σ_(court_id=5)(JUDGE_COURT)) ⋈_(judge_id = j.ID) ρ_j(JUDGE)
)
```

SQL:

```
SELECT j.ID, j.Name, j.Experience, j.Position
FROM Experience j
JOIN court_id jc ON j.ID = jc.judge_id
WHERE jc.court_id = 5;
```

Query 7: Cases Handled by a Specific Judge

Description: List all cases (case_no, name, status) handled by judge ID = 101

Relational Algebra:

```
π_(cs.case_no, cs.Name, cs.curr_status)(
  ((σ_(judge_id=101)(JUDGE_COURT)) ⋈_(court_id) COURT_CASE)
  ⋈_(case_no) ρ_cs(CASE)
)
```

SQL:

```
SELECT cs.case_no, cs.Name, cs.curr_status, cs.Category
FROM curr_status cs
JOIN case_no cn ON cs.case_no = cn.case_no
JOIN court_id jc ON cn.court_id = jc.court_id
WHERE jc.judge_id = 101;
```

Query 8: Lawyers Registered in Specific Bar Council

Description: List all lawyers (Name, BAR*Registration*No) registered in 'Gujarat' bar

Relational Algebra:

```
π_(Name, BAR_Registration_No, Type)(
  σ_(BAR_Registration_No LIKE 'GJ%')(LAWYER)
)
```

SQL:

```
SELECT ID, Name, BAR_Registration_No, Type
FROM BAR_Registration_No
WHERE BAR_Registration_No LIKE 'GJ%';
```

Query 9: Dormant Cases with Reasons

Description: List all dormant cases (case*no, name, dormant*reason)

Relational Algebra:

```
π_(cs.case_no, cs.Name, dr.dormant_reason)(
  (σ_(is_dormant=TRUE)(DORMANT_REASON)) ⋈_(ID = case_no) ρ_cs(CASE)
)
```

SQL:

```
SELECT cs.case_no, cs.Name, cs.Category, dr.dormant_reason, dr.curr_stage
FROM curr_status cs
JOIN dormant_reason dr ON cs.case_no = dr.ID
WHERE dr.is_dormant = TRUE;
```
